

Review Article: Sleep and Memory Consolidation

Cognitive Neuroscience - term paper

Introduction

Decades of research show that sleep does more than rest the body: it actively stabilizes and reorganizes memories formed during the day. This review summarizes evidence for the role of different sleep stages.

Slow-wave sleep and declarative memory

During deep non-REM sleep, the hippocampus replays recent experiences in compressed form. These replay events coincide with sleep spindles and slow oscillations in the cortex, and are thought to transfer facts and events into long-term cortical storage. Participants who sleep after learning word pairs recall more of them than those who stay awake for the same period.

REM sleep and emotional and procedural memory

REM sleep appears to support skill learning and the processing of emotional experiences. Some studies suggest REM strips the emotional charge from memories while keeping content.

Sleep deprivation

A single night without sleep reduces the ability to form new memories by roughly 40 percent, because the hippocampus cannot encode new information efficiently.

Targeted memory reactivation

Playing sounds or odors during slow-wave sleep that were present during learning can selectively strengthen those memories.

Practical implications

Students cramming all night trade sleep for study time and remember less. Short naps containing spindle-rich sleep can also improve recall.

Limitations

Many studies use small samples and laboratory tasks; real-world learning is less studied.